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Embedded systems

Keypad and Seven-Segment Display System

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1. Problem Statement

In many digital systems, a simple and reliable way of entering values is needed. A keypad is one of the most practical input devices because it is inexpensive, easy to use, and suitable for embedded applications. In this project, the keypad is used to input numbers and selected letters, while a seven-segment display is used to show the entered value immediately.

The challenge addressed in this work is to design a small embedded system that can read key presses, interpret them correctly, store consecutive inputs, and present them on a display. The system must support multi-digit entry such as 23 or 456. It should also recognize the # key as a reset command that clears the current display state. In addition, the letters A, B, C, and D must be displayed when their corresponding keys are pressed.

This project demonstrates the basic relationship between input handling, data processing, and output control in an embedded environment. It is a useful model for digital locks, calculators, counters, menu systems, and other interactive electronic devices.

2. Objectives of the Project

The main objective of the project is to design and simulate a keypad-controlled seven-segment display system in Tinkercad using Arduino. The system should interpret user input correctly and convert it into visible output on the display.

The specific objectives are to read keypad inputs in real time, display numbers and the characters A, B, C, and D, support consecutive multi-digit input, and clear the display when the # key is pressed. Another goal is to understand how embedded software and hardware work together to provide a responsive user interface.

3. Input and Output Components

The system is built from a small set of input and output components connected to an Arduino Uno, which acts as the controller.

Component Type	Description
Input Components	4×4 Keypad: Used to enter digits and the characters A, B, C, and D.
Input Components	# Key: Used to reset or clear the current display content.
Output Components	Seven-Segment Display: Shows the entered

	values visually.
Control Unit	Arduino Uno: Reads the keypad, processes the data, and drives the display.

The keypad provides the user input in a row-column matrix format. When a key is pressed, the Arduino identifies the row and column combination and translates it into the corresponding character. The display receives the output through segment control signals, with each segment turning on or off to form the required symbol.

4. System Design and Working Principle

The keypad and seven-segment display system follows a straightforward embedded control process. First, the Arduino continuously scans the keypad for any pressed key. When a key is detected, the controller compares the row and column position with the keypad map and identifies the character associated with the key.

If the pressed key is a digit, it is appended to the existing input buffer so that the system can show multi-digit values. For example, if 2 is pressed followed by 3, the stored sequence becomes 23. In the same way, 456 is formed by pressing 4, 5, and 6 consecutively. This buffering step is important because it allows the system to behave like a real digital input interface rather than showing only one key at a time.

If one of the alphanumeric keys A, B, C, or D is pressed, the corresponding character is shown according to the seven-segment encoding pattern used in the program. The # key functions as a clear command. When it is pressed, the buffer is erased and the output returns to an empty state. This makes the system easy to reset during testing or use.

The seven-segment display works by controlling seven individual LEDs named a, b, c, d, e, f, and g. Different combinations of these segments create different numbers and letters. Because one seven-segment digit can only show one symbol at a time, multi-digit output requires either multiple displays or a multiplexing technique. In the simulation, the display is updated continuously so that the user experiences a stable visual result.

5. Implementation in Tinkercad

The project was simulated in Tinkercad Circuits using an Arduino Uno, a 4×4 keypad, and a seven-segment display arrangement. The Arduino pins were assigned to the keypad rows and

columns for input scanning, while separate pins were connected to the display segments for output control. Current-limiting resistors were used to protect the display LEDs.

The program was written in the Arduino programming language. It includes a keypad library for easier key detection, a character-to-segment lookup table for display patterns, and a small memory buffer for storing the entered sequence. The code processes each key press, updates the buffer, and refreshes the display according to the stored value. During simulation, the system responded correctly to key presses. Numeric input appeared in sequence, the characters A, B, C, and D were displayed using the available segment patterns, and pressing # cleared the current output. The result shows that the design meets the intended requirements of the project.

6. Advantages and Limitations

One advantage of this system is simplicity. It uses a small number of components and is easy to understand, build, and test. It also demonstrates important embedded concepts such as scanning, buffering, encoding, and output refresh. Another advantage is flexibility, because the system can be expanded later to include a buzzer, password entry, menu navigation, or counting functions. The main limitation is that a seven-segment display has a restricted character set. Digits are easy to represent, but only a few letters can be shown clearly. Another limitation is that multi-digit output is more complex because it requires multiple displays or multiplexing. Despite these limits, the system is still very effective for basic embedded interfaces.

7. Conclusion

This project successfully demonstrates the design of a keypad-controlled seven-segment display system for an embedded environment. It shows how user input can be captured, processed, and displayed in real time using an Arduino Uno. The system supports multi-digit numeric input, displays the characters A, B, C, and D, and uses the # key as a reset command.

Through this work, the fundamental ideas of embedded systems were applied in a practical way. The project combines hardware wiring, software logic, and real-time response into one complete application. It is a strong foundation for more advanced systems that require reliable user interaction and visual feedback.

Reference

<https://www.tinkercad.com/things/jyRCg3gBreV/editel?returnTo=%2Fdashboard%2Fdesigns%2Fall>